

```
root@srv1: ~
root@srv1:~# apt list --installed | grep zfs
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

libzfs4linux/noble-updates,now 2.2.2-0ubuntu9.4 amd64 [installed,automatic]
zfs-zed/noble-updates,now 2.2.2-0ubuntu9.4 amd64 [installed,automatic]
zfsutils-linux/noble-updates,now 2.2.2-0ubuntu9.4 amd64 [installed]
root@srv1:~# zpool list
no pools available
root@srv1:~# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                8:0      0   30G  0 disk
├─sda1                            8:1      0    1M  0 part
├─sda2                            8:2      0    2G  0 part /boot
├─sda3                            8:3      0   28G  0 part
│   └─ubuntu--vg-ubuntu--lv      252:8     0    8G  0 lvm  /
│   └─ubuntu--vg-home--lv        252:9     0    2G  0 lvm  /home
sdb                                8:16     0    1G  0 disk
sdc                                8:32     0    1G  0 disk
sdd                                8:48     0    1G  0 disk
sde                                8:64     0    1G  0 disk
sdf                                8:80     0    1G  0 disk
sdg                                8:96     0    1G  0 disk
sdh                                8:112    0    1G  0 disk
sdi                                8:128    0    1G  0 disk
sdj                                8:144    0    1G  0 disk
sdk                                8:160    0   10G  0 disk
sdl                                8:176    0    2G  0 disk
├─vg_var-lv_var_rmeta_0          252:0     0    4M  0 lvm
│   └─vg_var-lv_var              252:4     0   1.1G  0 lvm  /var
├─vg_var-lv_var_rimage_0         252:1     0   1.1G  0 lvm
│   └─vg_var-lv_var              252:4     0   1.1G  0 lvm  /var
sdm                                8:192    0    2G  0 disk
├─vg_var-lv_var_rmeta_1          252:2     0    4M  0 lvm
│   └─vg_var-lv_var              252:4     0   1.1G  0 lvm  /var
├─vg_var-lv_var_rimage_1         252:3     0   1.1G  0 lvm
│   └─vg_var-lv_var              252:4     0   1.1G  0 lvm  /var
sr0                                11:0     1 1024M  0 rom
root@srv1:~# zpool create test_pool1 mirror /dev/sdb /dev/sdc
root@srv1:~# zpool create test_pool2 mirror /dev/sdd /dev/sde
root@srv1:~# zpool create test_pool3 mirror /dev/sdf /dev/sdg
root@srv1:~# zpool create test_pool4 mirror /dev/sdh /dev/sdi
root@srv1:~# zpool list
NAME      SIZE  ALLOC   FREE  CKPOINT  EXPANDSZ   FRAG    CAP  DEDUP    HEALTH  ALTROOT
test_pool1 960M   108K   960M      -         -         0%    0%  1.00x   ONLINE   -
test_pool2 960M   108K   960M      -         -         0%    0%  1.00x   ONLINE   -
test_pool3 960M   111K   960M      -         -         0%    0%  1.00x   ONLINE   -
test_pool4 960M   111K   960M      -         -         0%    0%  1.00x   ONLINE   -
```



```
root@srv1: ~  
NAME      SIZE  ALLOC  FREE  CKPOINT  EXPANDSZ  FRAG    CAP  DEDUP  HEALTH  ALTROOT  
test_pool1 960M  108K  960M    -        -        0%    0%  1.00x  ONLINE  -  
test_pool2 960M  108K  960M    -        -        0%    0%  1.00x  ONLINE  -  
test_pool3 960M  111K  960M    -        -        0%    0%  1.00x  ONLINE  -  
test_pool4 960M  111K  960M    -        -        0%    0%  1.00x  ONLINE  -  
root@srv1:~# zfs set compression=lzjb test_pool1  
root@srv1:~# zfs set compression=lz4 test_pool2  
root@srv1:~# zfs set compression=gzip-9 test_pool3  
root@srv1:~# zfs set compression=zle test_pool4  
root@srv1:~# zfs get all | grep compression  
test_pool1  compression      lzjb                local  
test_pool2  compression      lz4                 local  
test_pool3  compression      gzip-9              local  
test_pool4  compression      zle                 local  
root@srv1:~# ls /  
bin          dev  lib64          mnt  root          snap  test_pool1  tmp  
bin usr-is-merged etc  lib usr-is-merged opt  run          srv  test_pool2  usr  
boot         home lost+found     proc sbin          swap.img test_pool3  var  
cdrom        lib  media          raid sbin usr-is-merged sys  test_pool4  
root@srv1:~# for i in {1..4}; do wget -P /test_pool$i https://gutenberg.org/cache/epub/2600/pg2600.converter.log  
; done  
--2026-07-17 21:56:43-- https://gutenberg.org/cache/epub/2600/pg2600.converter.log  
Resolving gutenberg.org (gutenberg.org)... 152.19.134.47, 2610:28:3090:3000:0:bad:cafe:47  
Connecting to gutenberg.org (gutenberg.org)|152.19.134.47|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 41250559 (39M) [text/plain]  
Saving to: '/test_pool1/pg2600.converter.log'  
  
pg2600.converter.log      100%[=====>] 39.34M  2.20MB/s   in 19s  
  
2026-07-17 21:57:03 (2.04 MB/s) - '/test_pool1/pg2600.converter.log' saved [41250559/41250559]  
--2026-07-17 21:57:03-- https://gutenberg.org/cache/epub/2600/pg2600.converter.log  
Resolving gutenberg.org (gutenberg.org)... 152.19.134.47, 2610:28:3090:3000:0:bad:cafe:47  
Connecting to gutenberg.org (gutenberg.org)|152.19.134.47|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 41250559 (39M) [text/plain]  
Saving to: '/test_pool2/pg2600.converter.log'  
  
pg2600.converter.log      100%[=====>] 39.34M  2.22MB/s   in 31s  
  
2026-07-17 21:57:35 (1.26 MB/s) - '/test_pool2/pg2600.converter.log' saved [41250559/41250559]  
--2026-07-17 21:57:35-- https://gutenberg.org/cache/epub/2600/pg2600.converter.log  
Resolving gutenberg.org (gutenberg.org)... 152.19.134.47, 2610:28:3090:3000:0:bad:cafe:47  
Connecting to gutenberg.org (gutenberg.org)|152.19.134.47|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 41250559 (39M) [text/plain]
```



```
root@srv1: ~
Connecting to gutenber.org (gutenber.org)|152.19.134.47|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 41250559 (39M) [text/plain]
Saving to: '/test_pool3/pg2600.converter.log'

pg2600.converter.log      100%[=====>]  39.34M  3.14MB/s   in 11s

2026-07-17 21:57:47 (3.54 MB/s) - '/test_pool3/pg2600.converter.log' saved [41250559/41250559]

--2026-07-17 21:57:47--  https://gutenber.org/cache/epub/2600/pg2600.converter.log
Resolving gutenber.org (gutenber.org)... 152.19.134.47, 2610:28:3090:3000:0:bad:cafe:47
Connecting to gutenber.org (gutenber.org)|152.19.134.47|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 41250559 (39M) [text/plain]
Saving to: '/test_pool4/pg2600.converter.log'

pg2600.converter.log      100%[=====>]  39.34M  5.69MB/s   in 7.5s

2026-07-17 21:57:56 (5.26 MB/s) - '/test_pool4/pg2600.converter.log' saved [41250559/41250559]

root@srv1:~# ls -l /test_pool*
/test_pool1:
total 22128
-rw-r--r-- 1 root root 41250559 Jul  2 07:31 pg2600.converter.log

/test_pool2:
total 18021
-rw-r--r-- 1 root root 41250559 Jul  2 07:31 pg2600.converter.log

/test_pool3:
total 10973
-rw-r--r-- 1 root root 41250559 Jul  2 07:31 pg2600.converter.log

/test_pool4:
total 40312
-rw-r--r-- 1 root root 41250559 Jul  2 07:31 pg2600.converter.log
root@srv1:~# zfs list
NAME          USED  AVAIL  REFER  MOUNTPOINT
test_pool1    21.8M   810M   21.6M   /test_pool1
test_pool2    17.7M   814M   17.6M   /test_pool2
test_pool3    10.9M   821M   10.7M   /test_pool3
test_pool4    39.5M   792M   39.4M   /test_pool4
root@srv1:~# zfs get all | grep compressratio | grep -v ref
test_pool1    compressratio    1.82x    -
test_pool2    compressratio    2.23x    -
test_pool3    compressratio    3.66x    -
test_pool4    compressratio    1.00x    -
root@srv1:~#
```

```
root@srv1: ~  
root@srv1:~# pwd  
/root  
root@srv1:~# wget -O archive.tar.gz --no-check-certificate 'https://drive.userco  
ntent.google.com/download?id=1MvrceEp-WgAQe57aDEzxSRalPAwbNN1Bb&export=download'  
--2026-07-18 19:44:04-- https://drive.usercontent.google.com/download?id=1Mvrce  
p-WgAQe57aDEzxSRalPAwbNN1Bb&export=download  
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 142.251  
.38.97, 2a00:1450:400f:806::2001  
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)|142.25  
1.38.97|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 7275140 (6.9M) [application/octet-stream]  
Saving to: 'archive.tar.gz'  
  
archive.tar.gz      100%[=====>]    6.94M  7.98MB/s    in 0.9s  
  
2026-07-18 19:44:11 (7.98 MB/s) - 'archive.tar.gz' saved [7275140/7275140]  
  
root@srv1:~# tar -xzf archive.tar.gz  
zpoolexport/  
zpoolexport/filea  
zpoolexport/fileb  
root@srv1:~# ls -l  
total 7112  
-rw-r--r-- 1 root root 7275140 Dec  6  2023 archive.tar.gz  
drwxr-xr-x 2 root root   4096 May 15  2020 zpoolexport  
root@srv1:~# ls -l ./zpoolexport/  
total 1024008  
-rw-r--r-- 1 root root 524288000 May 15  2020 filea  
-rw-r--r-- 1 root root 524288000 May 15  2020 fileb  
root@srv1:~# zpool import -d zpoolexport/  
pool: otus  
id: 6554193320433390805  
state: ONLINE  
status: Some supported features are not enabled on the pool.  
(Note that they may be intentionally disabled if the  
'compatibility' property is set.)  
action: The pool can be imported using its name or numeric identifier, though  
some features will not be available without an explicit 'zpool upgrade'.  
config:  
  
otus                                ONLINE  
mirror-0                            ONLINE  
  /root/zpoolexport/filea          ONLINE  
  /root/zpoolexport/fileb          ONLINE  
root@srv1:~# zpool import -d zpoolexport/ otus
```



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root@srv1: ~  
/root/zpoolexport/fileb ONLINE  
root@srv1:~# zpool import -d zpoolexport/ otus  
root@srv1:~# zpool status  
pool: otus  
state: ONLINE  
status: Some supported and requested features are not enabled on the pool.  
The pool can still be used, but some features are unavailable.  
action: Enable all features using 'zpool upgrade'. Once this is done,  
the pool may no longer be accessible by software that does not support  
the features. See zpool-features(7) for details.  
config:  


| NAME                    | STATE  | READ | WRITE | CKSUM |
|-------------------------|--------|------|-------|-------|
| otus                    | ONLINE | 0    | 0     | 0     |
| mirror-0                | ONLINE | 0    | 0     | 0     |
| /root/zpoolexport/filea | ONLINE | 0    | 0     | 0     |
| /root/zpoolexport/fileb | ONLINE | 0    | 0     | 0     |

  
errors: No known data errors  
  
pool: test_pool1  
state: ONLINE  
config:  


| NAME       | STATE  | READ | WRITE | CKSUM |
|------------|--------|------|-------|-------|
| test_pool1 | ONLINE | 0    | 0     | 0     |
| mirror-0   | ONLINE | 0    | 0     | 0     |
| sdb        | ONLINE | 0    | 0     | 0     |
| sdc        | ONLINE | 0    | 0     | 0     |

  
errors: No known data errors  
  
pool: test_pool2  
state: ONLINE  
config:  


| NAME       | STATE  | READ | WRITE | CKSUM |
|------------|--------|------|-------|-------|
| test_pool2 | ONLINE | 0    | 0     | 0     |
| mirror-0   | ONLINE | 0    | 0     | 0     |
| sdd        | ONLINE | 0    | 0     | 0     |
| sde        | ONLINE | 0    | 0     | 0     |

  
errors: No known data errors  
  
pool: test_pool3  
state: ONLINE
```

```
root@srv1: ~  
pool: test_pool3  
state: ONLINE  
config:  


| NAME       | STATE  | READ | WRITE | CKSUM |
|------------|--------|------|-------|-------|
| test_pool3 | ONLINE | 0    | 0     | 0     |
| mirror-0   | ONLINE | 0    | 0     | 0     |
| sdf        | ONLINE | 0    | 0     | 0     |
| sdg        | ONLINE | 0    | 0     | 0     |

  
errors: No known data errors  
  
pool: test_pool4  
state: ONLINE  
config:  


| NAME       | STATE  | READ | WRITE | CKSUM |
|------------|--------|------|-------|-------|
| test_pool4 | ONLINE | 0    | 0     | 0     |
| mirror-0   | ONLINE | 0    | 0     | 0     |
| sdh        | ONLINE | 0    | 0     | 0     |
| sdi        | ONLINE | 0    | 0     | 0     |

  
errors: No known data errors  
root@srv1:~# zfs get available otus  
NAME PROPERTY VALUE SOURCE  
otus available 350M -  
root@srv1:~# zfs get readonly otus  
NAME PROPERTY VALUE SOURCE  
otus readonly off default  
root@srv1:~# zfs get recordsize otus  
NAME PROPERTY VALUE SOURCE  
otus recordsize 128K local  
root@srv1:~# zfs get compression otus  
NAME PROPERTY VALUE SOURCE  
otus compression zle local  
root@srv1:~# zfs get checksum otus  
bad property list: invalid property 'checksum'  
usage:  
get [-rHp] [-d max] [-o "all" | field[,...]]  
[-t type[,...]] [-s source[,...]]  
<"all" | property[,...]> [filesystem|volume|snapshot|bookmark] ...  
  
The following properties are supported:  


| PROPERTY | EDIT | INHERIT | VALUES |
|----------|------|---------|--------|
|----------|------|---------|--------|


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root@srv1: ~  
root@srv1:~# zfs get checksum otus  
NAME PROPERTY VALUE SOURCE  
otus checksum sha256 local  
root@srv1:~# wget -O otus_task2.file --no-check-certificate https://drive.userco  
ntent.google.com/download?id=lwgxjih8YZ-cqLqaZVa0lA3h3Y029c3oI&export=download  
[1] 2698  
root@srv1:~#  
Redirecting output to 'wget-log'.  
^C  
[1]+ Done wget -O otus_task2.file --no-check-certificate ht  
ps://drive.usercontent.google.com/download?id=lwgxjih8YZ-cqLqaZVa0lA3h3Y029c3oI  
root@srv1:~# ls  
archive.tar.gz otus_task2.file wget-log zpoollexport  
root@srv1:~# rm otus_task2.file  
root@srv1:~# wget -O otus_task2.file --no-check-certificate 'https://drive.userc  
ontent.google.com/download?id=lwgxjih8YZ-cqLqaZVa0lA3h3Y029c3oI&export=download'  
--2026-07-18 20:02:52-- https://drive.usercontent.google.com/download?id=lwgxji  
h8YZ-cqLqaZVa0lA3h3Y029c3oI&export=download  
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 142.251  
.38.97, 2a00:1450:400f:806::2001  
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)|142.25  
1.38.97|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 5432736 (5.2M) [application/octet-stream]  
Saving to: 'otus_task2.file'  
  
otus_task2.file 100%[=====>] 5.18M 7.63MB/s in 0.7s  
  
2026-07-18 20:02:55 (7.63 MB/s) - 'otus_task2.file' saved [5432736/5432736]  
  
root@srv1:~# ls  
archive.tar.gz otus_task2.file wget-log zpoollexport  
root@srv1:~# zfs receive otus/test@today < otus_task2.file  
root@srv1:~# ls /otus  
hometask2 test  
root@srv1:~# ls /otus/test  
10M.file for_examaple.txt Limbo.txt task1 world.sql  
cinderella.tar homework4.txt Moby_Dick.txt War_and_Peace.txt  
root@srv1:~# find /otus/test -name "secret_message"  
/otus/test/task1/file_mess/secret_message  
/otus/test/task1/file_mess/secret_message  
-bash: /otus/test/task1/file_mess/secret_message: Permission denied  
root@srv1:~# cat /otus/test/task1/file_mess/secret_message  
https://otus.ru/lessons/linux-hl/  
  
root@srv1:~#
```